

Context: GenAI vs. good old thinking

Two issues with Generative AI (GenAI) chatbots are:

🌿 **Environmental cost:** they rely on power-hungry data centres ($\approx 1\%$ of global electricity today), and a single AI query can consume multiple times the energy of a typical search

🧠 **Judgement erosion:** they encourage *cognitive offloading*, where users rely on automated answers rather than engaging in deeper active reasoning, reinforcing *automation bias* (the tendency to accept outputs uncritically)

*Rubber ducking** is a method of solving a problem[†] by articulating it in spoken or written natural language whereby the experience of explaining the issue results in finding/realising the solution

[†] often debugging, but not exclusively!

The hard work is the *prompting* - which might render the actual GenAI *query* redundant!



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AI folks have now discovered “thinking”

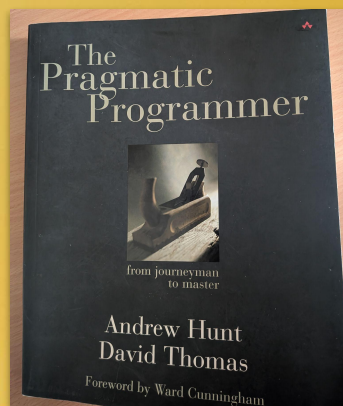


Steph Smith ✓

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Sometimes in the process of writing a good enough prompt for ChatGPT, I end up solving my own problem, without even needing to submit it.

How could we promote ‘rubber ducking’?



*as canonised by this iconic book on software engineering (here's my trusty copy!)

💡 **Means to broadcast widely the value of ‘rubber ducking’** (e.g. green, cognitive and career) as opposed to blindly executing queries to a GenAI ?

💡 Create and demonstrate a **placebo tool** to illustrate the concept (*ChatGPlaceboT*) ? Same UI but this (green) tool only returns static probing questions e.g. *what assumptions are you making, what conditions did you see this in, etc.* Suggested **best/green practice for the prompt engineering stage** ? E.g. intercept to ‘think’:



(ChatG)PlaceboT:
do you need an AI chatbot or just a rubber duck?

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